MATHS MANTHAN
• TUITION CLASSES •
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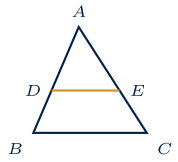
Subject: Mathematics | **PERIODIC COMPREHENSIVE TEST – 07** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Marks are indicated against each question.
2. **Section A** comprises 6 questions of 1 mark each (5 MCQs and 1 Assertion-Reason).
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based integrated unit of assessment of 4 marks.
7. Use of calculators is strictly prohibited. Draw neat, labeled geometric diagrams wherever required.

SECTION A: OBJECTIVE QUESTIONS (6 Marks)



1. In the given figure, if $\angle ADE = \angle B$, with $AD = 3.8$ cm, $AE = 3.6$ cm, $BE = 2.1$ cm, and $BC = 4.2$ cm, then DE is equal to:

(a) 2.8 cm (b) 2.1 cm (c) 3.2 cm (d) 1.9 cm

2. In ΔABC and ΔDEF, $\angle B = \angle E$, $\angle F = \angle C$, and $AB = 3DE$. Then the two triangles are:

(a) Congruent but not similar (b) Similar but not congruent (c) Neither congruent nor similar (d) Congruent and similar

3. If D, E, F are respectively the midpoints of sides BC, CA, AB of ΔABC, then ΔDEF is similar to:

(a) ΔABC (b) Only ΔBDF (c) Only ΔCDE (d) None of these

4. In ΔABC, D and E are points on sides AB and AC such that $DE \parallel BC$. If $AD = 4x - 3$, $DB = 3x - 1$, $AE = 8x - 7$, and $EC = 5x - 3$, the value of x is:

(a) 1 (b) 2 (c) $-\frac{1}{2}$ (d) 3

5. A scale model of a cargo ship is made to a scale of 1 : 200. If the length of the model is 4 m, the actual length of the ship is:

(a) 800 m (b) 50 m (c) 80 m (d) 204 m

6. **Assertion (A):** Diagonals of a trapezium divide each other proportionally.
Reason (R): The triangles formed by the diagonals and parallel sides of a trapezium are similar by AA criterion.

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

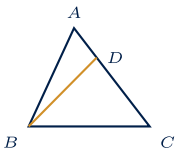
SECTION B: VERY SHORT ANSWER QUESTIONS (4 Marks)

7. **[2 Marks]** In ΔABC, D and E are points on sides AB and AC such that $DE \parallel BC$. If $AD = 1.5$ cm, $DB = 3$ cm, and $BC = 6$ cm, find the length of line segment DE.

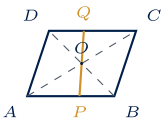
8. **[2 Marks]** If $\Delta ABC \sim \Delta PQR$ with $\angle A = 48°$ and $\angle B = 72°$, determine the exact degree measure of $\angle R$.

SECTION C: SHORT ANSWER QUESTIONS (6 Marks)

9. **[3 Marks]** In the given figure, D is a point on side AC of ΔABC such that $\angle ABD = \angle C$. If $AB = 8$ cm and $AC = 16$ cm, calculate the length of segment AD.



10. **[3 Marks]** The diagonals of a parallelogram $ABCD$ intersect at O. A line through O intersects side AB at P and side CD at Q. Prove that line segment $OP = OQ$.



SECTION D: LONG ANSWER / HOTS QUESTIONS (10 Marks)

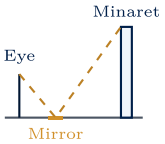
11. **[5 Marks] State and prove Basic Proportionality Theorem (Thales' Theorem).**
Application: In trapezium $ABCD$ with $AB \parallel DC$, a line EF parallel to AB is drawn intersecting AD at E and BC at F. Using Thales' Theorem, prove that:

$$\frac{AE}{ED} = \frac{BF}{FC}$$

12. **[5 Marks] State Converse of Basic Proportionality Theorem.**
In ΔABC, D and E are points on sides AB and AC respectively such that $\frac{AD}{DB} = \frac{AE}{EC}$ and $\angle ADE = \angle ACB$. Prove that ΔABC is an isosceles triangle.

SECTION E: COMPETENCY / CASE STUDY QUESTION (4 Marks)

13. **Case Study 07: Mirror Reflection Triangulation of an Ancient Minaret**



To determine the height of an ancient minaret, an archaeology team places a flat mirror horizontally on level ground 18 m away from the base of the minaret. A researcher stands 1.5 m behind the mirror such that he can see the reflection of the pinnacle of the minaret in the center of the mirror. The researcher's eye level is 1.8 m above the ground.

(i) **[1 Mark]** Which physical law of optics guarantees that the two triangles formed are equiangular and similar?

(ii) **[1 Mark]** Name the geometric similarity criterion applicable between the two triangles.

(iii) **[2 Marks]** Calculate the exact height of the ancient minaret above the horizontal ground.

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★